NUMERICAL ANALYSIS
PROBLEM SET 2

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We begin by noting that all of the files referenced in this document are available at:
https://github.com/chtompki/MathSpring2026/tree/main/NumericalAnalysis/ProblemSet2

1. Use the command `magic(n)`, for $n = 100, 200, 300, \ldots, 10^4$ to construct the magic square matrices of order n. Magic squares are defined as those matrices having entries for which the sum of the elements by rows, columns, or diagonals are identical. Then compute their determinants by the command `det()`. Use the command `cputime` to get the CPU time that is needed fore this computation. **Be patient, it may take around 10 minutes to compute these determinants.**

Find a constant C such that the computation time t_n scales approximately as t_n Cn^3. Plot your data and approximate scaling using `semilogy()`. Think back to the methods you learned for computing determinants in your Linear Algebra class. What method do you suspect Matlab is using and why? How can you be sure it isn't using cofactor expansion?

Solution. We begin by noticing that for any $A =$ `magic(n)` where n is even, $\det(A) = 0$. So we are less concerned with keeping the valued of the determinants, and are more interested in the number of seconds that it takes to calculate the i^{th}-determinant. A sample of the code (given in file ps2prob1.m) that we are using follows as:

```
determinants=zeros(1,100);
cpuTime=zeros(1,100);
for n=100:100:10000
    A = magic(n);
    tstart = cputime;
    det(A);
    cpuTime(n/100) = cputime - tstart;
end
plotN=1:100;
figure;hold on; semilogyi(plotN, 0.000008.*plotN.^3, '--x', plotN, cpuTime, '-x');
xlabel('Matrix Size (n/100)');
ylabel('Value');
```

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```
legend('Cn^3','CpuTime');
title('Determinants and CPU Time vs. Matrix Size');
```

Notice that around the determinant calculation we are taking the system time and doing a subtraction to get the duration of the determinant algorithm. The figure that we get from plotting the cputime follows in Figure 1 Lastly we notice that for $C = 2.5 \times 10^{-5}$, which we

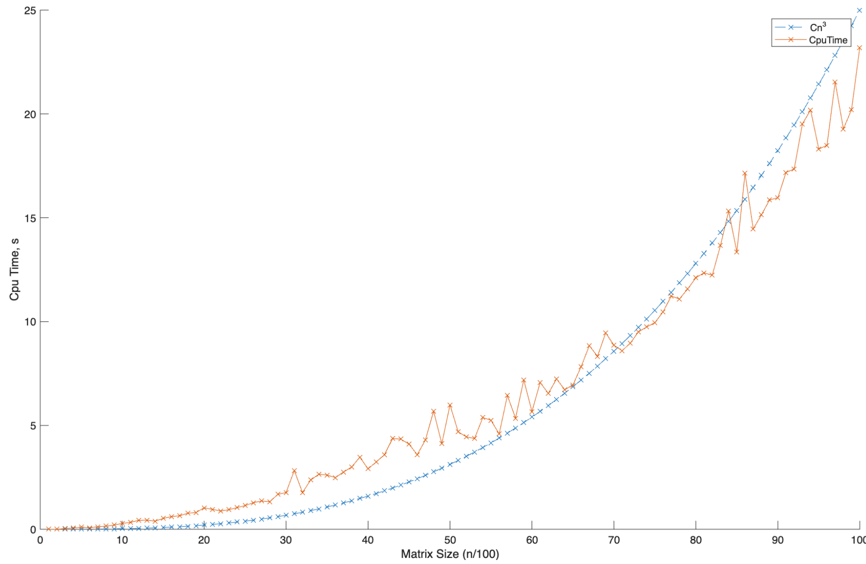

FIGURE 1. Determinant Runtime by Matrix Size

found by fiddling around with the algorithm to get the value using the bisector method by hand so to speak. □

2. Consider the matrix $A = \begin{pmatrix} 201 & 199 \\ 199 & 201 \end{pmatrix}$.

(a) Compute the spectral radius of $\rho(A)$.
(b) Computer the condition number $K(A)$. You may choose any natural matrix norm, but be sure to specify which one you are using.
(c) Assume some numerical method has been applied to obtain the approximate solution $\hat{x} = (-0.5, 0.5)^\top$ to the system $Ax = B$ for $b = (-2, 2)^\top$. Use the residual $r = A\hat{x} - b$ and known inequalities to provide a bound on the norm of the relative error $\frac{\|\hat{x}-x\|}{\|x\|}$.
(d) Find the exact solution to the above system (you may use any method you prefer, but show you work)., Compare the relative error estimate from above to the actual relative error.
(e) In general, when can you expect the residual r to provide a tight bound on the error?

Solution. For the problem generally we'll use the L_2-norm.

(a) To compute the spectral radius $\rho(A)$, we go after the eigenvalues of A. So, let's compute the characteristic polynomial.

$$p(t) = \begin{vmatrix} 201-t & 199 \\ 199 & 201-t \end{vmatrix} = t^2 - 402t - 40401 - 39601 = t^2 - 402t + 800.$$

So we solve for $p(t) = 0$, and we get the following roots λ_1, λ_2 from the quadratic formula:

$$\lambda_1, \lambda_2 = \frac{402 \pm \sqrt{161604 - 3200}}{2} = \frac{402 \pm 398}{2} = 2, 400.$$

So for (a), our spectral radius follows as:

$$\rho(A) = \max\{\lambda_1, \lambda_2\} = 400.$$

(b) We'll continue to use the L_2-norm here. Notice that $A \in \text{symm}(\mathbb{R})$, so we know that the eigenvalues λ_1, λ_2 set up as:

$$\|A\|_2 = 400 \quad \text{and} \quad \|A^{-1}\|_2 = \frac{1}{2}.$$

So by definition of $K(A)$, we have

$$K_2(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{400}{2} = 200.$$

(c) We start with some matrix multiplication to see what our vectors $\hat{x}, b$ from the problem statement give us. So we see that

$$\begin{bmatrix} 201 & 199 \\ 199 & 201 \end{bmatrix} \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 201(-0.5) + 199(0.5) \\ 199(-0.5) + 201(0.5) \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

So write the residual as

$$r = A\hat{x} - b = \begin{bmatrix} -1 \\ 1 \end{bmatrix} - \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

So we let the error be $e = \hat{x} - x$. Notice

$$r = A\hat{x} - b = A\hat{x} - Ax = A(\hat{x} - x) = Ae.$$

We can use our condition number theorem for this bit of the problem now where we notice that

$$\frac{\|e\|}{\|x\|} \leq K(A) \frac{\|r\|}{\|b\|}.$$

We already have $K(A)$, so we only need calculate $\|r\|/\|b\|$. But we can do a quick calculation on those values as:

$$\|r\| = \sqrt{1^2 + (-1)^2} = \sqrt{2} \quad \text{and} \quad \|b\| = \sqrt{((2)^2 + (-2)^2)} = 2\sqrt{2}$$

So we have that
$$\frac{\|\hat{x}-x\|}{\|x\|} \leq 200 \cdot \frac{1}{2} = 100.$$

(d) If we compute the actual error what do we get? Begin by noticing that
$$e = \hat{x} - x = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix} - \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}$$
So we can simply now compute $\|e\| = 1/\sqrt{2}$, and $\|x\| = \sqrt{2}$, so the actual error is
$$\frac{\|\hat{x}-x\|}{\|x\|} = \frac{1}{2}$$

(e) We go back to our theorem:
$$\frac{\|e\|}{\|x\|} \leq K(A)\frac{\|r\|}{\|b\|}.$$
For our bound to be tight, we need the inequalities above to be close to equalities. And for these equalities to be close we need that $K(A)$ be well conditioned and $\|r\|/\|b\|$

□

3. Consider the linear system $Ax = b$ with
$$A = \begin{bmatrix} 2 & -2 & 0 \\ \epsilon - 2 & 2 & 0 \\ 0 & -1 & 3 \end{bmatrix}$$

(a) Compute the LU factorization of the given A by hand using Gaussian Elimination, and show that the matrix L has the element $\ell_{32} = -1/\epsilon$.

(b) Compute the LU factorization by hand using the Crout algorithm, and show the matrix L has the element $\ell_{22} = \epsilon$ in this case.

(c) Compute the condition number $K(A)$. You may use any norm you like, but be sure to specify which one you use.

(d) Multiply Ax for $x = [111]^\top$ to find the corresponding b for which $x = [111]^\top$ is a solution to $Ax = b$. Your answer should depend upon ϵ. For values $\epsilon = 10^{-k}$, with $k = 0, \ldots, 9$ numerically implement the Crout method of LU decomposition to solve the system $Ax = b$. Compute the relative error as a function of ϵ. Compare your results based on the Crout algorithm to results obtained using the build in `lu(A)` function.

(e) Repeat the process above for $x = [\log(2.5) \quad 1 \quad 1]^\top$ for values $\epsilon = 1/3 \times 10^{-k}$, $k = 0, \ldots, 9$. Compare the relative error for this choice of b to the previous case. Why didn't you see the impact of roundoff error in the previous case, but you do in this case? Discuss your numerical error results for this case in the context of theoretical error estimates based on your computation of the condition number $K(A)$.

Solution. Whoops I did a full Gauss Jordan elimination and left it instead of deleting it. We begin with our augmented matrix:

$$\left[\begin{array}{ccc|ccc} 2 & -2 & 0 & 1 & 0 & 0 \\ \epsilon-2 & 2 & 0 & 0 & 1 & 0 \\ 0 & -1 & 3 & 0 & 0 & 1 \end{array}\right]$$

Perform the row operation $r2 : (-\epsilon/2+1)r_1 + r_2$, where ":" denotes replacement, seen as follows

$$\left[\begin{array}{ccc|ccc} 2 & -2 & 0 & 1 & 0 & 0 \\ 0 & \epsilon & 0 & \frac{-\epsilon}{2}+1 & 1 & 0 \\ 0 & -1 & 3 & 0 & 0 & 1 \end{array}\right]$$

Let's now get rid of ϵ on the left hand side of our matrix by the row substitution $r_2 : r_2/\epsilon$, giving us

$$\left[\begin{array}{ccc|ccc} 2 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{-1}{2}+\frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ 0 & -1 & 3 & 0 & 0 & 1 \end{array}\right]$$

Let's now look at making the leading entry in the matrix one, by $r_1 : r_1/2$ as:

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & \frac{-1}{2}+\frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ 0 & -1 & 3 & 0 & 0 & 1 \end{array}\right].$$

Perform $r_1 : r_2 + r_1$ as

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ 0 & 1 & 0 & \frac{-1}{2}+\frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ 0 & -1 & 3 & 0 & 0 & 1 \end{array}\right].$$

Similarly, perform $r_3 : r_2 + r_3$ given by

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ 0 & 1 & 0 & \frac{-1}{2}+\frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ 0 & 0 & 3 & -\frac{1}{2}+\frac{1}{\epsilon} & \frac{1}{\epsilon} & 1 \end{array}\right].$$

Lastly perform $r_3 : r_3/3$ as:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ 0 & 1 & 0 & \frac{-1}{2}+\frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ 0 & 0 & 1 & -\frac{1}{6}+\frac{1}{3\epsilon} & \frac{1}{3\epsilon} & \frac{1}{3} \end{array}\right].$$

Well dang it, I wrote all that out and I know that it's not *LU* decomposition. We'll have another go, see below.

(a) We begin with our augmented matrix with the identity on the left hand side:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -2 & 0 \\ 0 & 1 & 0 & \epsilon-2 & 2 & 0 \\ 0 & 0 & 1 & 0 & -1 & 3 \end{array}\right].$$

We want to clear the element at 2×1, so we perform the row operation $r_2 : (-\epsilon/2+1)r_1 + r2$ which gives us

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -2 & 0 \\ \frac{-\epsilon}{2}+1 & 1 & 0 & 0 & \epsilon & 0 \\ 0 & 0 & 1 & 0 & -1 & 3 \end{array}\right].$$

To finish off our LU Gaussian factorization, we perform the row operation $r_3 : (-1/\epsilon)r_2 + r_3$ as

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -2 & 0 \\ \frac{-\epsilon}{2}+1 & 1 & 0 & 0 & \epsilon & 0 \\ \frac{1}{2}-\frac{1}{\epsilon} & -\frac{1}{\epsilon} & 1 & 0 & 0 & 3 \end{array}\right],$$

which as stipulated has $\ell_{32} = -1/\epsilon$.

(b) The Crout decomposition is an iterative process with a goal to result in the following decomposition

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} \ell_{11} & 0 & 0 \\ \ell_{21} & \ell_{22} & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}.$$

which we do by approaching the matrix by its rows, moving from the top down. We begin by having the following

$$\ell_{11} = a_{11} = 2 \quad \text{and} \quad \ell_{21} = a_{21} = \epsilon - 2 \quad \text{and} \quad \ell_{31} = a_{31} = 0.$$

Next we go after ℓ_{12} and ℓ_{13} for our $j = 1$ step, in the following manner

$$\ell_{12} = \frac{a_{12}}{\ell_{12}} = -\frac{2}{2} = -1.\ell_{13} = \frac{a_{13}}{\ell_{13}} = \frac{0}{2} = 0$$

We then move to the $j = 2$ step and compute ℓ_{22} and ℓ_{32}

$$\begin{aligned} \ell_{22} &= a_{22} - \ell_{21}u_{12} = 2 - (\epsilon-2)(-1) = 2 + (\epsilon - 2) = \epsilon \\ \ell_{32} &= a_{32} - \ell_{31}u_{12} = -1 - 0(-1) = -1 \end{aligned}$$

Lastly for $j = 3$ we have

$$\ell_{33} = a_{33} - (\ell_{31}u_{13} + \ell_{32}u_{23}) = 3 - (0 \cdot 0)$$

So we see that

$$A = \begin{bmatrix} 2 & 0 & 0 \\ \epsilon - 2 & \epsilon 0 & \\ 0 & -\frac{1}{\epsilon} & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

as the LUdecomposition for the matrix.

(c) To compute the conditinon number $K(A)$, we're going to use the L_1-norm so that $K(A) = \|A\|_1 \|A^{-1}\|_1$. The column sums of A follow as

- Column 1: $|2| + |\epsilon - 2| + 0 = 2 + |\epsilon - 2|$
- Column 2: $|-2| + |2| + |-1| = 5$
- Column 3: 3

So $\|A\|_1 = \max\{2+|\epsilon-2|, 3, 5\}$, which for $\epsilon > 0$ small we have $\|A\|_1 = \max\{2+|\epsilon-2|, 3, 5\} = 5$.

We have, out of shear luck, have already calculated A^{-1} by accident in (a) as

$$A^{-1} = \begin{bmatrix} \frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ \frac{-1}{2} + \frac{1}{\epsilon} & \frac{1}{\epsilon} & 0 \\ -\frac{1}{6} + \frac{1}{3\epsilon} & \frac{1}{3\epsilon} & \frac{1}{3} \end{bmatrix}.$$

We to compute $\|A^{-1}\|_1$, like the first portion of (c), go about computing the column sums of A^{-1} as

- Column 1: $1/\epsilon - 1/2 + 1/\epsilon - 1/6 + 1/(3\epsilon) = -2/3 + 7/(3\epsilon)$
- Column 2: $1/\epsilon + 1/\epsilon + 1/(3\epsilon) = 7/(3\epsilon)$
- Column 3: $1/(3\epsilon)$.

Thus, we have

$$\|A^{-1}\|_1 = \max\{-\frac{2}{3} + \frac{7}{3\epsilon}, \frac{7}{3\epsilon}, \frac{1}{3\epsilon}\}$$

For $0 < \epsilon \leq 1$, $\|A^{-1}\|_1 = 7/(3\epsilon)$. So,

$$K(A) = \|A\|_1 \|A^{-1}\|_1 = 5 \cdot \frac{7}{3\epsilon} \qquad (0 < \epsilon \leq 1),$$

which has $K(A)$ blows up as $1/\epsilon$.

(d) Now I don't know if there's a typographical error in the problem statement such that what is given as x should have been given as b. We have worked the problem as best we could, given the statement, if I understand correctly. We do the algebraic expansion of $b = Ax$ with $x = [1 \quad 1 \quad 1]^\top$ as

$$b = \begin{bmatrix} 2 & -2 & 0 \\ \epsilon - 2 & 2 & 0 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 - 2 + 0 \\ (\epsilon - 2) + 2 \\ -1 + 3 \end{bmatrix} = \begin{bmatrix} 0 \\ \epsilon \\ 2 \end{bmatrix}$$

We are confused here with the problem, because it asks to "solve" the equation $Ax = b$ when we have been given A and x. This makes it seem as if we are to perform a computation of a multiplication as opposed to solving to try to find a solution. To complete the problem we set $\epsilon = 10^{-k}$. This yields

$$b = \begin{bmatrix} 0 \\ 10^{-k} \\ 2 \end{bmatrix}$$

which we can compute for a variety of k's. For example, we have

$$b = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ \frac{1}{10} \\ 2 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ \frac{1}{10^9} \\ 2 \end{bmatrix},$$

How we're to use this to do a comparison on Gauss Jordan eludes us. We lastly note that b converges to the vector

$$b = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}.$$

(e) We can repeat the same process in (d) with the vector $x = [\log(2.5) \quad 1 \quad 1]^\top$. Our computation for b becomes

$$b = \begin{bmatrix} 2 & -2 & 0 \\ \epsilon - 2 & 2 & 0 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} \log(2.5) \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2\log(2.5) - 2 + 0 \\ (\epsilon - 2)\log(2.5) + 2 \\ -1 + 3 \end{bmatrix} = \begin{bmatrix} 2\log(2.5) - 2 \\ (\epsilon - 2)\log(2.5) + 2 \\ 2 \end{bmatrix}.$$

Now we set $\epsilon = 1/3 \times 10^{-k}$ and run $k = 0, \dots, 9$. The solutions again fall out as:

$$\begin{aligned} b \quad = \quad & \begin{bmatrix} 2\log(2.5) - 2 \\ (\frac{1}{3} - 2)\log(2.5) + 2 \\ 2 \end{bmatrix}, \\ & \begin{bmatrix} 2\log(2.5) - 2 \\ (\frac{1}{30} - 2)\log(2.5) + 2 \\ 2 \end{bmatrix}, \dots, \\ & \begin{bmatrix} 2\log(2.5) - 2 \\ (\frac{1}{3\times 10^9} - 2)\log(2.5) + 2 \\ 2 \end{bmatrix}. \end{aligned}$$

Which gives us a collection of b vectors that converge on the following vector:

$$b = \begin{bmatrix} 2\log(2.5) - 2 \\ (-2)\log(2.5) + 2 \\ 2 \end{bmatrix}$$

Lastly, we know that the condition number $K(A) = 5 \cdot \frac{7}{3\epsilon}$ makes sense because all of our solutions b vary in the disappearance of a term.

Note. We wrote a Crout LU solver while fiddling with this problem which is committed lucrout.m □

4. Consider the matrix as follows

$$A = \begin{bmatrix} 1 & 1 + 0.5 \times 10^{-15} & 3 \\ 2 & 2 & 20 \\ 3 & 6 & 4 \end{bmatrix}.$$

Compute the LU decomposition by Crout's method (without pivoting). You may carry out the procedure in Matlab, but show your code and the output after iterating through each stage. Compute the residual $R = A - LU$. Compare the result to the residual for LUP decomposition obtained with pivoting by using the Matlab command `[L U] = lu(A)` and computing $R_P = PA - LU$.

Solution. In working the solution for problem 3 we developed a Crout algorithm which we've already referred to. The algorithm code follows as

```
function [L,U] = lucrout(A)
    n = size(A,1);
    if size(A,2) ~= n
        error('A must be square');
    end
    L = zeros(n,n);
    U = eye(n);
    for j = 1:n
        for i = j:n
            L(i,j) = A(i,j) - L(i,1:j-1)*U(1:j-1,j);
        end
        if L(j,j) == 0
            error('Zero pivot at j=%d (need pivoting).', j);
        end
        for k = j+1:n
```

```
            U(j,k) = (A(j,k) - L(j,1:j-1)*U(1:j-1,k)) / L(j,j);
        end
    end
end
```

The file for this code is: lucrout.m, with the solution for the problem at ps2prob4.m,

When we did the decomposition on A from the problem statement, we ended up with

$$A = \left(1.0 \times 10^{16} \begin{bmatrix} 0.000000000000000 & 0 & 0 \\ 0.000000000000000 & -0.000000000000000 & 0 \\ 0.000000000000000 & 0.000000000000000 & 4.728779608739018 \end{bmatrix} \right) \cdot \left(1.0 \times 10^{16} \begin{bmatrix} 0.000000000000000 & 0.000000000000000 & 0.000000000000000 \\ 0 & 0.000000000000000 & -1.576259869579674 \\ 0 & 0 & 0.000000000000000 \end{bmatrix} \right)$$

which clearly suffers from roundoff error because of the lack of conditioning of the matrix A. When we ran $R = A - LU$, we ended up with

$$R = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{bmatrix} \neq [0]$$

Though, when we went about using the Matlab library function `[L, U, P] = lu(A)`, we ended up with

$$A = \begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 1 & 0 \\ \frac{1}{3} & \frac{1}{2} & 0 \end{bmatrix} \cdot \begin{bmatrix} 3 & 6 & 4 \\ 0 & -2 & \frac{52}{3} \\ 0 & 0 & -7 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

And if we compute the residual $R_P = PA - LU$, we end up with:

$$R_P = \begin{bmatrix} 2 & 5 & 1 \\ 0 & 0 & 0 \\ -2 & -5 & 3 \end{bmatrix}.$$

$\square$

5. Consider the following matrix:

$$A = \begin{bmatrix} 1 & 1-\epsilon & 3 \\ 2 & 2 & 2 \\ 3 & 6 & 4 \end{bmatrix}.$$

(a) Find the values of ε for which LU decomposition (without pivoting) is not possible.

(b) Find the LU decomposition for the value of ε for which A is singular.
(c) What in the LU decomposition signals that A is singular for this case?

Solution. Before we discuss (a), (b), or (c), we go to Matlab to perform the decomposition using `[L, U, P]=lu(A)` given as:

$$A = \begin{bmatrix} 1 & 1-\epsilon & 3 \\ 2 & 2 & 2 \\ 3 & 6 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & \frac{3}{2} - \frac{3}{2\epsilon} & 1 \end{bmatrix} \begin{bmatrix} 1 & 1-\epsilon & 3 \\ 0 & 2\epsilon & -4 \\ 0 & 0 & 1+\frac{6}{\epsilon} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(a) When we send $\epsilon \to 0$ we have $\ell_{22} = 2\epsilon = 0$.

(b) If we want to have A singular we can set $u_{33} = 1 + \frac{6}{\epsilon} = 0$ so that the dimension of the row space being at most 2 by the third row being all zeros. Solving our u_{33} equation for ϵ gives $\epsilon = -6$.

(c) The key for a LU decomposition to be singular is to consider the non-one's diagonal of either L or U depending on the decomposition algorithm. If any of these non-ones values is zero, then the whole decomposition is singular because the determinant of the original matrix is zero. □

6. Show that for a symmetric positive definite matrix A we have $K_2(A^2) = (K_2(A))^2$.
Hint. Make use of the relation between the minimum and maximum eigenvalues $\lambda_{min}, \lambda_{max}$ and condition number $K_2(A)$ associated to the L^2 natrual matrix norm for a symmetrix positive definite matrix A.

Proof. We'll perform the proof relying on the L_2-norm. Note that if A is symmetric positive definite, then we have a matricies Q and Λ such that $A = Q\Lambda Q^\top$, where $\Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_i > 0$. For symmetric matrices the singular values equal $|lambda_i|$. Now notice,

$$\|A\|_2 = \mathrm{max}_i \lambda_i = \lambda_{\max} \quad \text{and} \quad \|A^{-1}\|_2 = \mathrm{max}_i \frac{1}{\lambda_i} = \frac{1}{\lambda_{\min}}.$$

Thus we have,

$$K_2(A) = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

Now we want to compute A^2 to see what we get. So consider

$$A^2 = (Q\lambda Q^\top)(Q\lambda Q^\top) = Q\lambda^2 Q^\top$$

So the eigenvalues of Λ^2 are λ_i^2 which are all positive. Thus, we see that,

$$\|A^2\|_2 = \lambda_{\max}^2, \qquad \|(A^2)^{-1}\|_2 = \|(A^{-1})^2\|_2 = \frac{1}{\lambda_{\min^2}}.$$

And, so

$$K_2(A^2) = \|A^2\|_2 \|(A^2)^{-1}\|_2 = \lambda_{\max}^2 \frac{1}{\lambda_{\min^2}} = \left(\frac{\lambda_{\max}}{\lambda_{\min}}\right)^2 = (K(A))^2,$$

and we are done. □

7. Prove that the following are symmetric positive definite matrices. Implement the Cholesky decomposition algorithm and use it to compute the Cholesky decomposition for each.

$$A = \begin{bmatrix} 16 & -28 & 0 \\ -28 & 53 & 10 \\ 0 & 10 & 29 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & -2 & 3 & -2 \\ -2 & 20 & -2 & 8 \\ 3 & -2 & 11 & -5 \\ -2 & 8 & -5 & 9 \end{bmatrix}.$$

Solution. Note we have code for the solution of the problem available at: ps2prob7.m

Consider the arbitrary $\mathbb{R}^3$ vector $q = [x \quad y \quad z]^\top$. We'll now epand $x^\top Ax$, calling it $Q(x)$ for convenience as

$$Q(x,y,z) = x^\top Ax = 53y^3 - 28xy + 10yz + 16x^2 - 28xy + 10yz + 29z^2 = 16x^2 - 56xy + 53y^2 + 20yz + 29z^2.$$

We'll need to do some algebraic manipulation to get what we are looking for:

$$\begin{aligned} 16x^2 - 56xy &= 16\left(x^2 - \frac{56}{10}xy\right) = 16(x^2 - \frac{7}{2}xy) \\ x^2 - \frac{7}{2}xy &= \left(x - \frac{7}{4}y\right)^2 - \frac{49}{16}y^2 \end{aligned}$$

Substituting this all back into Q we see that

$$Q = 16\left(x - \frac{7}{4}y\right)^2 + 4y^2 + 20yz + 29z^2.$$

Now we complete the square in y, z. Consider $4y^2 + 20xy - 29z^2 = 4(y^2 - 5xy) + 29z^2$. A little manipulation gets us to

$$624y^2 + 20xy + 29z^2 = 4\left(y\frac{5}{2}z\right) + 4z^2$$

So we can finally construct

$$Q(x,y,z) = 16\left(x - \frac{7}{4}y)^2\right) + 4\left(y + \frac{5}{2}z\right)^2 + 4z^2.$$

This is the sum of squared terms so it can't be zero or negative.

Another technique for showing a matrix symmetric positive definite is using Leading Principal Submatrix factorization. The leading principal factors follow as

- $\Delta_1 = b_{11} = 1 > 0$
- $\Delta_2 = \det \begin{bmatrix} 1 & -2 \\ -2 & 20 \end{bmatrix} = 16 > 0$
- $\Delta_3 = \det \begin{bmatrix} 1 & -2 & 3 \\ -2 & 20 & -2 \\ 3 & -2 & 11 \end{bmatrix} = 16 > 0$
- $\Delta_4 = \det(B) = 64 > 0$

So we are positive definite.

We've added in the algorithm for the Cholesky method at: cholesky.m The resulting matricies L_A, and L_B, computed using the Cholesky method and Matlab follow as:

$$L_A = \begin{bmatrix} 4 & 0 & 0 \\ -7 & 2 & 0 \\ 0 & 5 & 2 \end{bmatrix}, \qquad L_B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 4 & 0 & 0 \\ 3 & 1 & 1 & 0 \\ -2 & 1 & 0 & 2 \end{bmatrix}.$$

$\square$